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United States Patent	12402518
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Cao; Weiran et al.

Display panel with reduced gas breaking and foreign matter invasion and manufacturing method thereof

Abstract

A display panel and a manufacturing method thereof are provided. A light-emitting device layer is disposed on a substrate, and an encapsulation layer covers the light-emitting device layer and the substrate. The encapsulation layer includes a first inorganic layer, a getter layer, an organic layer, and a second inorganic layer, and the first inorganic layer covers the light-emitting device layer, wherein the getter layer is disposed on the first inorganic layer, the organic layer is disposed on the getter layer, the second inorganic layer covers the organic layer and the getter layer and is connected to the first inorganic layer.

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Appl. No.:	17/622843
Filed (or PCT Filed):	December 21, 2021
PCT No.:	PCT/CN2021/140089
PCT Pub. No.:	WO2023/108719
PCT Pub. Date:	June 22, 2023

Prior Publication Data

Foreign Application Priority Data

CN202111535128.4Dec. 15, 2021

Publication Classification

Int. Cl.: H10K59/80 (20230101); H10K59/12 (20230101)

U.S. Cl.:

CPC H10K59/874 (20230201); H10K59/1201 (20230201);

Field of Classification Search

CPC: H10K (59/874); H10K (59/1201); H10K (59/873); H10K (2102/331)

USPC: 257/91

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6740145	12/2003	Boroson	257/E23.137	F26B 21/083
2006/0284556	12/2005	Tremel	313/506	H10K 59/874
2007/0013305	12/2006	Wang	313/553	C03C 8/02
2008/0042547	12/2007	Chien	313/311	H01J 7/18
2010/0136239	12/2009	Wang	252/181.7	B01J 20/28033
2014/0368107	12/2013	Jung	313/504	H10K 50/846
2016/0088756	12/2015	Ramadas	N/A	N/A
2017/0027168	12/2016	Heath	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1894790	12/2010	CN	N/A
104241320	12/2013	CN	N/A
104704035	12/2014	CN	N/A
106653517	12/2016	CN	N/A
106848092	12/2016	CN	N/A
107492601	12/2016	CN	N/A
109046248	12/2017	CN	N/A
109904345	12/2018	CN	N/A
109994524	12/2018	CN	N/A
110459694	12/2018	CN	N/A
111129349	12/2019	CN	N/A
214226948	12/2020	CN	N/A

OTHER PUBLICATIONS

International Search Report in International application No. PCT/CN2021/140089, mailed on Aug. 25, 2022. cited by applicant

Written Opinion of the International Search Authority in International application No. PCT/CN2021/140089, mailed on Aug. 25, 2022. cited by applicant

Chinese Office Action issued in corresponding Chinese Patent Application No. 202111535128.4 dated Apr. 29, 2023, pp. 1-10. cited by applicant

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Background/Summary**FIELD OF INVENTION**

(1) This application relates to a field of display technology and particularly relates to a display panel and a manufacturing method thereof.

BACKGROUND OF INVENTION

(2) In a process of research and practice of prior art, the inventor of this application found that common undesirable black spots in current mass-produced organic light-emitting diode panels seriously affect product quality. Through analyzing many cases where an inorganic encapsulation layer is broken due to foreign matter, it is found that the main reason is that gas released by an organic encapsulation layer and external water vapor invaded from cracks of a film layer, thus damaging a cathode and organic light-emitting diode (OLED) materials, and forming dot-like luminescence abnormalities.

Technical Problem

(3) Embodiments of the present application provide a display panel and a manufacturing method thereof, which can absorb the gas released by the organic layer in the encapsulation layer and reduce the risk of occurrence of undesirable black spots.

SUMMARY OF INVENTION

(4) An embodiment of the present application provides a display panel, which includes: a substrate; a light-emitting device layer disposed on the substrate; and at least one encapsulation layer, wherein the encapsulation layer includes: a first inorganic layer covering the light-emitting device layer; a getter layer disposed on the first inorganic layer; an organic layer disposed on the gettering layer; and a second inorganic layer covering the organic layer and the getter layer and connected to the first inorganic layer; wherein a material of the getter layer includes getter graphene, the getter graphene is formed of graphene-based carbon material and getter nanoparticle through chemical groups bonding, and the graphene-based carbon materials are bonded to each other through the chemical groups.

(5) Optionally, in some embodiments of the present application, the graphene-based carbon material includes at least one graphene or carbon nanotube.

(6) Optionally, in some embodiments of the present application, the getter layer further includes a resin and a photo initiator.

(7) Optionally, in some embodiments of the present application, the getter nanoparticle has a porous structure.

(8) Optionally, in some embodiments of the present application, the first inorganic layer includes a

first boundary, the getter layer includes a second boundary, the organic layer includes a third boundary, and the second inorganic layer includes a fourth boundary, wherein the third boundary is defined on an inner side of the second boundary, and the second boundary is defined on an inner side of the first boundary; and wherein the second inorganic layer is respectively connected to the second boundary and the third boundary, and the fourth boundary is located outside the first boundary or coincides with the first boundary.

(9) Optionally, in some embodiments of the present application, a distance from the first boundary to the second boundary is greater than a distance from the second boundary to the third boundary.

(10) Optionally, in some embodiments of the present application, a groove is defined on a side of the gettering layer facing the organic layer, and the organic layer fills the groove.

(11) Optionally, in some embodiments of the present application, the getter nanoparticles are molecular sieves or nano metal organic framework molecules.

(12) An embodiment of the present application further provides a display panel, which includes: a substrate; a light-emitting device layer disposed on the substrate; and at least one encapsulation layer, wherein the encapsulation layer includes: a first inorganic layer covering the light-emitting device layer; a getter layer disposed on the first inorganic layer; an organic layer disposed on the gettering layer; and a second inorganic layer covering the organic layer and the getter layer and connected to the first inorganic layer.

(13) Optionally, in some embodiments of the present application, a material of the getter layer comprises getter graphene, the getter graphene is formed of graphene-based carbon material and getter nanoparticle through chemical groups bonding.

(14) Optionally, in some embodiments of the present application, the graphene-based carbon materials are bonded to each other through the chemical groups.

(15) Optionally, in some embodiments of the present application, a groove is defined on a side of the gettering layer facing the organic layer, and the organic layer fills the groove.

(16) Optionally, in some embodiments of the present application, the graphene-based carbon material includes at least one graphene or carbon nanotube.

(17) Optionally, in some embodiments of the present application, the getter layer further includes a resin and a photo initiator.

(18) Optionally, in some embodiments of the present application, each getter nanoparticle has a porous structure.

(19) Optionally, in some embodiments of the present application, the first inorganic layer includes a first boundary, the getter layer includes a second boundary, the organic layer includes a third boundary, and the second inorganic layer includes a fourth boundary, wherein the third boundary is defined on an inner side of the second boundary, and the second boundary is defined on an inner side of the first boundary; and wherein the second inorganic layer is respectively connected to the second boundary and the third boundary, and the fourth boundary is located outside the first boundary or coincides with the first boundary.

(20) Optionally, in some embodiments of the present application, a distance from the first boundary to the second boundary is greater than a distance from the second boundary to the third boundary.

(21) Correspondingly, this application further relates to a manufacturing method of a display panel, which includes the following steps: forming a light-emitting device layer on a substrate; forming at least one encapsulation layer on the light-emitting device layer; wherein the step of forming the at least one encapsulation layer on the light-emitting device layer includes the following steps: forming a first inorganic layer on the light-emitting device layer; forming a getter layer on the first inorganic layer; forming an organic layer on the gettering layer; and forming a second inorganic layer on the organic layer, wherein the second inorganic layer covers the organic layer and the getter layer and is connected to the first inorganic layer.

(22) Optionally, in some embodiments of the present application, the step of forming the getter layer on the first inorganic layer includes the following steps: bonding graphene-based carbon

materials and getter nanoparticles through chemical groups to form getter graphene; mixing the getter graphene, resin, and photo initiator to form a mixed material; applying the mixed material to the first inorganic layer to form a mixed material layer; and curing the mixed material layer to form the gettering layer.

(23) Optionally, in some embodiments of the present application, the graphene-based carbon material includes at least one graphene or carbon nanotube.

(24) In the display panel in the embodiment of the present application, the light-emitting device layer is disposed on the substrate, the encapsulation layer covers the light-emitting device layer and the substrate. The encapsulation layer includes a first inorganic layer, a getter layer, an organic layer, and a second inorganic layer. The first inorganic layer covers the light-emitting device layer. The getter layer is disposed on the first inorganic layer. The organic layer is arranged on the getter layer. The second inorganic layer covers the organic layer and the getter layer and is connected to the first inorganic layer.

BENEFICIAL EFFECT

(25) In the display panel and the manufacturing method thereof in the embodiments of the present application, a getter layer is formed in the encapsulation layer to absorb the gas released by the organic layer, thereby reducing the amount of gas in the encapsulation layer. This reduces the risk of gas breaking the inorganic layer, and prevents the first inorganic layer from being damaged by foreign matter and causing external water and oxygen to invade the light-emitting device layer.

Description

DESCRIPTION OF DRAWINGS

(1) In order to describe the technical solutions in the embodiments of the present application more clearly, the following will briefly introduce the drawings that need to be used in the description of the embodiments. Obviously, the drawings in the following description are only some embodiments of the application. For those skilled in the art, other drawings can be obtained based on these drawings without doing creative work.

(2) FIG. 1 is a schematic structural diagram of a display panel provided by an embodiment of the present application.

(3) FIG. 2 is a schematic structural diagram of getter graphene in the display panel provided by an embodiment of the present application.

(4) FIG. 3 is a schematic flow chart of a manufacturing method of a display panel provided by an embodiment of the present application.

(5) FIG. 4 is a schematic structural diagram of step B1 in the manufacturing method of the display panel provided by an embodiment of the present application.

(6) FIG. 5 is a schematic structural diagram of step B2 in the manufacturing method of the display panel provided by an embodiment of the present application.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(7) The following will describe the technical solutions in the embodiments of the present application clearly and completely with reference to the drawings in the embodiments of the present application. Obviously, the embodiments are only a part of the embodiments of the present application, rather than all the embodiments. Based on these embodiments in this application, all other embodiments obtained by those skilled in the art without doing creative work shall fall within the protection scope of this application. In addition, the specific implementations described herein are only used to illustrate and explain the present application, and are not used to limit the present application. In this application, if no explanation is made to the contrary, the orientation terms such as “upper” and “lower” usually refer to the upper and lower positions of the device in actual use or working state. Specifically, it is the direction in the drawings, and “inner” and “outer” refer to the

outline of the device.

(8) The embodiment of the present application provides a display panel and a manufacturing method thereof, which will be described in detail below. It should be noted that the order of description in the following embodiments is not meant to limit the preferred order of the embodiments.

(9) Please refer to FIG. 1, an embodiment of the present application provides a display panel **100**, which includes a substrate **11**, a light-emitting device layer **12**, and at least one encapsulation layer **13**.

(10) The light-emitting device layer **12** is disposed on the substrate **11**. The encapsulation layer **13** covers the light-emitting device layer **12** and the substrate **11**.

(11) The encapsulation layer **13** includes a first inorganic layer **131**, a getter layer **132**, an organic layer **133**, and a second inorganic layer **134**. The first inorganic layer **131** covers the light-emitting device layer **12**. The getter layer **132** is disposed on the first inorganic layer **131**. The organic layer **133** is disposed on the getter layer **132**. The second inorganic layer **134** covers the organic layer **133** and the getter layer **132** and is connected to the first inorganic layer **131**.

(12) In the display panel **100** in the embodiment of the present application, a getter layer **132** is formed in the encapsulation layer **13** to absorb the gas released by the organic layer **133**, which reduces the amount of gas in the encapsulation layer **13**, and reduces the risk of gas breaking the inorganic layer. In addition, this prevents external water and oxygen from invading the light-emitting device layer when the first inorganic layer is damaged by foreign matter, thereby reducing the formation of black spots.

(13) Optionally, substrate **11** includes a base and a thin film transistor disposed on the base. The substrate can be a rigid substrate or a flexible substrate. The material of the substrate includes any one of glass, sapphire, silicon, silicon dioxide, polyethylene, polypropylene, polystyrene, polylactic acid, polyethylene dicarboxylate, polyethylene terephthalate, polyethylene naphthalate, polycarbonate, polyethersulfone, aromatic fluorotoluene containing polyarylate, polycyclic olefin, polyimide, or polyurethane.

(14) Optionally, the light-emitting device layer **12** includes a first electrode, a light-emitting layer, and a second electrode that are sequentially disposed on the substrate **11**. The light-emitting device layer **12** may be a top emission type or a bottom emission type.

(15) That is, one of the first electrode and the second electrode is a reflective electrode, and the other is a transparent electrode.

(16) Optionally, the material of the light-emitting layer may be an organic material. For example, it can be Alq₃, bis (2-methyl-8-hydroxyquinoline-N1,O8)-(1,1'-biphenyl-4-hydroxy) aluminum (BALq), DPVBi, Almq₃, 3-tert-butyl-9,10-bis (2-naphthalene) anthracene (TBADN).

(17) The material of the light-emitting layer can also be an inorganic material. For example, it can be selected from at least one of group IV semiconductor nanocrystals, group II-V semiconductor nanocrystals, group II-VI semiconductor nanocrystals, group IV-VI semiconductor nanocrystals, group III-V semiconductor nanocrystalline, or group III-VI semiconductor nanocrystals. For example, it can be at least one of silicon quantum dots, germanium quantum dots, cadmium sulfide quantum dots, cadmium selenide quantum dots, cadmium telluride quantum dots, zinc selenide quantum dots, lead sulfide quantum dots, lead selenide quantum dots, indium phosphide quantum dots, indium arsenide quantum dots, or gallium nitride quantum dots.

(18) Optionally, the number of layers of the encapsulation layer **13** may be one or plural.

Optionally, the material of the first inorganic layer **131** and the second inorganic layer **134** may be selected from at least one of silicon oxide, silicon nitride, silicon oxynitride, aluminum oxide, or magnesium oxide.

(19) The material of the first inorganic layer **131** may also be other metal oxides. Optionally, the material of the getter layer **132** includes getter graphene xs. The getter graphene xs is formed of graphene-based carbon material sm and getter nanoparticle xq through chemical group bonding.

(20) In this embodiment, graphene is used as the material of the getter layer **132**. Because graphene has excellent water and oxygen barrier properties, the water and oxygen barrier performance of the getter layer **132** can be improved. In addition, graphene is bonded with getter nanoparticles, which improves the dispersibility of graphene. In the microstructure, a spacer structure of layered graphene and gettering nanoparticles is formed, which can effectively absorb the gas released by the organic layer **133**. Secondly, graphene has good ductility and can be used as a stress relief layer to reduce the shrinkage of the organic layer after curing and the damage to the inorganic layer caused by thermal expansion.

(21) Optionally, the chemical group may be an epoxy group, an ether group, or an oxygen-containing group such as an ester group, or a hydrogen-containing group or a fluorine-containing group.

(22) Wherein, the surface of the graphene-based carbon material **sm** may be connected to at least one getter nanoparticle **xq** through a chemical group. For example, the surface of the graphene-based carbon material **sm** can be connected to at least one getter nanoparticle **xq** through a chemical group.

(23) In addition, as shown in FIG. 2, one getter nanoparticle **xq** can also connect at least one graphene-based carbon material **sm** particle through a chemical group. For example, a getter nanoparticle **xq** bonds two graphene particles through two chemical groups.

(24) Graphene-based carbon materials **sm** can also be bonded together through chemical groups. For example, graphene particles are bonded to carbon nanotube particles through chemical groups, at least two graphene particles are bonded together through chemical groups, or at least two carbon nanotube particles are bonded together through chemical groups.

(25) Optionally, in some embodiments, the graphene-based carbon material **sm** includes graphene and carbon nanotubes. The graphene particles are bonded to the carbon nanotube particles, and the graphene particles are bonded to the getter nanoparticles **xq**.

(26) Optionally, the getter nanoparticle **xq** has a porous structure to improve the gas absorption performance of the getter layer **132**. In some embodiments, the getter nanoparticle **xq** may not be a porous material.

(27) Optionally, the getter nanoparticles **xq** may be particles such as molecular sieves or nano metal organic framework molecules.

(28) Optionally, the getter layer **132** further includes a resin **sz** and a photo initiator, that is, the getter graphene **xs**, resin, and photo initiator are mixed to form the getter layer **132**.

(29) Optionally, the resin **sz** may be epoxy resin or acrylic resin or the like.

(30) Optionally, in some embodiments, a groove is formed on the side of the getter layer **132** facing the organic layer **133**, and the organic layer **133** fills the grooves. The formation of grooves on the getter layer **132** improves the flexibility of the getter layer **132** on the one hand, and increases the contact area between the organic layer **133** and the getter layer **132** on the other hand. Furthermore, the efficiency of the getter layer **132** in absorbing the gas released by the organic layer is improved, and the stability of the connection between the getter layer **132** and the organic layer **133** is improved.

(31) Optionally, the viscosity of the material of the getter layer **132** ranges from 100 cps to 500 cps. For example, it can be 100 cps, 200 cps, 300 cps, 400 cps, or 500 cps. The thickness of the getter layer **132** is less than 5 micrometers. For example, it can be 4 micrometers, 3 micrometers, 2 micrometers, or 1 micrometer.

(32) Optionally, the first inorganic layer **131** includes a first boundary **13a**. The getter layer **132** includes a second boundary **13b**. The organic layer **133** includes a third boundary **13c**. The second inorganic layer **134** includes a fourth boundary **13d**. The third boundary **13c** is defined inside the second boundary **13b**. The second boundary **13b** is defined inside the first boundary **13a**.

(33) The second inorganic layer **134** is connected to the second boundary **13b** and the third boundary **13c**, respectively. The fourth boundary **13d** is located outside the first boundary **13a** or

coincides with the first boundary **13a**. Such an arrangement improves the stability of the encapsulation layer **13** so that the second inorganic layer **134** is connected to the first inorganic layer **131**, the getter layer **132**, and the organic layer **133**.

(34) Optionally, the distance from the first boundary **13a** to the second boundary **13b** is greater than the distance from the second boundary **13b** to the third boundary **13c**. This arrangement further improves the stability of the encapsulation layer **13**.

(35) Please refer to FIG. **3**. Correspondingly, the present application further relates to a manufacturing method of the display panel **100**, which includes the following steps: Step **B1**: forming a light-emitting device layer on a substrate; and Step **B2**: forming at least one encapsulation layer on the light-emitting device layer; wherein the step of forming the at least one encapsulation layer on the light-emitting device layer includes following steps: Step **B21**: forming a first inorganic layer on the light-emitting device layer; Step **B22**: forming a getter layer on the first inorganic layer; Step **B23**: forming an organic layer on the gettering layer; and Step **B24**: forming a second inorganic layer on the organic layer, wherein the second inorganic layer covers the organic layer and the getter layer and is connected to the first inorganic layer.

(36) In the manufacturing method of the display panel in the embodiment of the present application, a getter layer is formed in the encapsulation layer to absorb the gas released by the organic layer, which reduces the amount of gas in the encapsulation layer, thereby reducing the risk of gas breaking the inorganic layer.

(37) The manufacturing method of the display panel **100** will be described below. It should be noted that the manufacturing method is used to manufacture the display panel **100** of the aforesaid embodiment.

(38) Please refer to FIG. **4**. In step **B1**, a light-emitting device layer **12** is formed on the substrate **11**.

(39) Wherein, substrate **11** includes a base and a thin film transistor disposed on the base. The substrate can be a rigid substrate or a flexible substrate. The material of the substrate includes any one of glass, sapphire, silicon, silicon dioxide, polyethylene, polypropylene, polystyrene, polylactic acid, polyethylene dicarboxylate, polyethylene terephthalate, polyethylene naphthalate, polycarbonate, polyethersulfone, aromatic fluorotoluene containing polyarylate, polycyclic olefin, polyimide, or polyurethane.

(40) Optionally, the light-emitting device layer **12** includes a first electrode, a light-emitting layer, and a second electrode that are sequentially disposed on the substrate **11**. The light-emitting device layer **12** may be a top emission type or a bottom emission type.

(41) That is, one of the first electrode and the second electrode is a reflective electrode, and the other is a transparent electrode.

(42) Optionally, the material of the light-emitting layer may be an organic material. For example, it can be Alq₃, bis (2-methyl-8-hydroxyquinoline-N1,O8)-(1,1'-biphenyl-4-hydroxy) aluminum (BALq), DPVBi, Almq₃, 3-tert-butyl-9,10-bis (2-naphthalene)anthracene (TBADN).

(43) The material of the light-emitting layer can also be an inorganic material. For example, it can be selected from at least one of group IV semiconductor nanocrystals, group II-V semiconductor nanocrystals, group II-VI semiconductor nanocrystals, group IV-VI semiconductor nanocrystals, group III-V semiconductor nanocrystalline, or group III-VI semiconductor nanocrystals. For example, it can be at least one of silicon quantum dots, germanium quantum dots, cadmium sulfide quantum dots, cadmium selenide quantum dots, cadmium telluride quantum dots, zinc selenide quantum dots, lead sulfide quantum dots, lead selenide quantum dots, indium phosphide quantum dots, indium arsenide quantum dots, or gallium nitride quantum dots.

(44) Next, proceed to step **B2**.

(45) Please refer to FIG. **5**. In step **B2**, at least one encapsulation layer **13** is formed on the light-emitting device layer **12**.

(46) Step **B2** includes the following steps: Step **B21**: forming a first inorganic layer **131** on the

light-emitting device layer **12**.

(47) Optionally, the material of the first inorganic layer **131** may be selected from at least one of silicon oxide, silicon nitride, silicon oxynitride, aluminum oxide, or magnesium oxide. The material of the first inorganic layer **131** may also be other metal oxides. Next, proceed to step B22.

Step B22: forming a getter layer **132** on the first inorganic layer **131**.

(48) Wherein, step B22 includes the following steps: Step B221: The graphene-based carbon material sm and the getter nanoparticle xq are bonded through chemical groups to form getter graphene xs.

(49) Optionally, the graphene-based carbon material sm includes at least one of graphene or carbon nanotubes.

(50) Optionally, the chemical group may be an epoxy group, an ether group, or an oxygen-containing group such as an ester group, or a hydrogen-containing group or a fluorine-containing group.

(51) The surface of the graphene-based carbon material sm may be connected to at least one getter nanoparticle xq through a chemical group. For example, the surface of the graphene-based carbon material sm can be connected to at least one getter nanoparticle xq through a chemical group.

(52) In addition, as shown in FIG. 2, one getter nanoparticle xq can also connect at least one graphene-based carbon material sm particle through a chemical group. For example, a getter nanoparticle xq bonds two graphene particles through two chemical groups.

(53) Graphene-based carbon materials sm can also be bonded together through chemical groups. For example, graphene particles are bonded to carbon nanotube particles through chemical groups, at least two graphene particles are bonded together through chemical groups, or at least two carbon nanotube particles are bonded together through chemical groups.

(54) Optionally, in some embodiments, the graphene-based carbon material sm includes graphene and carbon nanotubes. The graphene particles are bonded to the carbon nanotube particles, and the graphene particles are bonded to the getter nanoparticles xq.

(55) Optionally, the getter nanoparticle xq has a porous structure to improve the gas absorption performance of the getter layer **132**. In some embodiments, the getter nanoparticle xq may not be a porous material.

(56) Optionally, the getter nanoparticles xq may be particles such as molecular sieves or nano metal organic framework molecules. Step B222: mixing the getter graphene xs, resin sz, and photo initiator to form a mixed material. The viscosity of the mixed material ranges from 100 cps to 500 cps, for example, it can be 100 cps, 200 cps, 300 cps, 400 cps, or 500 cps. Step B223: applying the mixed material to the first inorganic layer **131** to form a mixed material layer. The mixed material layer can be formed on the first inorganic layer **131** by spin coating or inkjet printing. Step B24: curing the mixed material layer to form the gettering layer **132**.

(57) The mixed material layer can be cured by ultraviolet light, and the mixed material layer can be baked to form the getter layer **132**. Step B23: forming an organic layer **133** on the gettering layer **132**. Step B24: forming a second inorganic layer **134** on the organic layer **133**. The second inorganic layer **134** covers the organic layer **133** and the getter layer **132** and is connected to the first inorganic layer **131**.

(58) Optionally, the material of the second inorganic layer **134** may be selected from at least one of silicon oxide, silicon nitride, silicon oxynitride, aluminum oxide, or magnesium oxide. The material of the second inorganic layer **134** may also be other metal oxides.

(59) In this way, the manufacturing process of the display panel **100** of the embodiment of the present application is completed.

(60) In the display panel and the manufacturing method thereof in the embodiments of the present application, the light-emitting device layer is disposed on the substrate, and the encapsulation layer covers the light-emitting device layer and the substrate. The encapsulation layer includes a first inorganic layer, a getter layer, an organic layer, and a second inorganic layer, and the first inorganic

layer covers the light-emitting device layer. The getter layer is disposed on the first inorganic layer, the organic layer is disposed on the getter layer, and the second inorganic layer covers the organic layer and the getter layer and is connected to the first inorganic layer.

(61) In the display panel and the manufacturing method thereof in the embodiments of the present application, a getter layer is formed in the encapsulation layer to absorb the gas released by the organic layer, which reduces the amount of gas in the encapsulation layer, thereby reducing the risk of gas breaking the inorganic layer.

(62) In the foregoing, the display panel and the manufacturing method thereof provided by the embodiments of the present application are described in detail. Specific examples are used in this article to illustrate the principles and implementation of the present application. The description of the above examples is only used to help understand the methods and core ideas of the present application. In addition, for those skilled in the art, according to the idea of the application, there will be changes in the specific implementation and the scope of the present application. As described above, the content of this specification should not be construed as limiting the present application.

Claims

1. A display panel comprising: a substrate; a light-emitting device layer disposed on the substrate; and at least one encapsulation layer, wherein the encapsulation layer comprises: a first inorganic layer covering the light-emitting device layer; a getter layer disposed on the first inorganic layer; an organic layer disposed on the gettering layer; and a second inorganic layer covering the organic layer and the getter layer and connected to the first inorganic layer; wherein a material of the getter layer comprises getter graphene, the getter graphene is formed of graphene-based carbon material and getter nanoparticles through chemical groups bonding, and the graphene-based carbon materials are bonded to each other through the chemical groups, wherein the first inorganic layer comprises a first boundary, the getter layer comprises a second boundary, the organic layer comprises a third boundary, and the second inorganic layer comprises a fourth boundary, wherein the third boundary is defined on an inner side of the second boundary, and the second boundary is defined on an inner side of the first boundary, and the second boundary and the third boundary are connected, and wherein a viscosity of the material of the getter layer ranges from 100 cps to 500 cps, a thickness of the getter layer is less than 5 micrometers, and the getter nanoparticles are molecular sieves or nano metal organic framework molecules.
2. The display panel of claim 1, wherein the graphene-based carbon material comprises at least one graphene or carbon nanotube.
3. The display panel of claim 1, wherein the getter layer further comprises a resin and a photo initiator.
4. The display panel of claim 1, wherein each getter nanoparticle has a porous structure.
5. The display panel of claim 1, wherein the second inorganic layer is respectively connected to the second boundary and the third boundary, and the fourth boundary is located outside the first boundary or coincides with the first boundary.
6. The display panel of claim 5, wherein a distance from the first boundary to the second boundary is greater than a distance from the second boundary to the third boundary.
7. The display panel of claim 1, wherein a groove is defined on a side of the gettering layer facing the organic layer, and the organic layer fills the groove.
8. A display panel, comprising: a substrate; a light-emitting device layer disposed on the substrate; and at least one encapsulation layer, wherein the encapsulation layer comprises: a first inorganic layer covering the light-emitting device layer; a getter layer disposed on the first inorganic layer; an organic layer disposed on the gettering layer; and a second inorganic layer covering the organic layer and the getter layer and connected to the first inorganic layer, wherein the first inorganic layer

comprises a first boundary, the getter layer comprises a second boundary, the organic layer comprises a third boundary, and the second inorganic layer comprises a fourth boundary, wherein the third boundary is defined on an inner side of the second boundary, and the second boundary is defined on an inner side of the first boundary, and the second boundary and the third boundary are connected, and wherein a viscosity of the material of the getter layer ranges from 100 cps to 500 cps, a thickness of the getter layer is less than 5 micrometers.

9. The display panel of claim 8, wherein a material of the getter layer comprises getter graphene, the getter graphene is formed of graphene-based carbon material and getter nanoparticles through chemical groups bonding.

10. A display panel, comprising: a substrate; a light-emitting device layer disposed on the substrate; and at least one encapsulation layer, wherein the encapsulation layer comprises: a first inorganic layer covering the light-emitting device layer; a getter layer disposed on the first inorganic layer; an organic layer disposed on the gettering layer; and a second inorganic layer covering the organic layer and the getter layer and connected to the first inorganic layer, wherein a material of the getter layer comprises getter graphene, the getter graphene is formed of graphene-based carbon material and getter nanoparticles through chemical groups bonding, wherein the first inorganic layer comprises a first boundary, the getter layer comprises a second boundary, the organic layer comprises a third boundary, and the second inorganic layer comprises a fourth boundary, wherein the third boundary is defined on an inner side of the second boundary, and the second boundary is defined on an inner side of the first boundary, and the second boundary and the third boundary are connected, and wherein a viscosity of the material of the getter layer ranges from 100 cps to 500 cps, a thickness of the getter layer is less than 5 micrometers, and the getter nanoparticles are molecular sieves or nano metal organic framework molecules.

11. The display panel of claim 10, wherein the graphene-based carbon material comprises at least one graphene or carbon nanotube.

12. The display panel of claim 10, wherein the getter layer further comprises a resin and a photo initiator.

13. The display panel of claim 10, wherein each getter nanoparticle has a porous structure.

14. The display panel of claim 10, wherein the second inorganic layer is respectively connected to the second boundary and the third boundary, and the fourth boundary is located outside the first boundary or coincides with the first boundary.

15. The display panel of claim 14, wherein a distance from the first boundary to the second boundary is greater than a distance from the second boundary to the third boundary.
